arXiv mixed with Tableau-Style Report

Faked using R+Quarto+LaTeX code hidden

for Stats 422

Abstract

This is the abstract. It should be a single paragraph summarizing the methodology and results. It will appear centered.

Introduction

We have reached the end. We hope that you have sufficient tools to write a report with the styling you desire. There are two versions of this (they are the same) the second one reveals the code for teaching purposes

Lorem ipsum filler

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Bar Chart

The humble bar chart/bar plot is one of the most effective tools in data visualization. Use them to compare values across different categories with rectangular bars where the length of the bar is proportional to the value it represents. Recall the human processes the meaning most correctly.

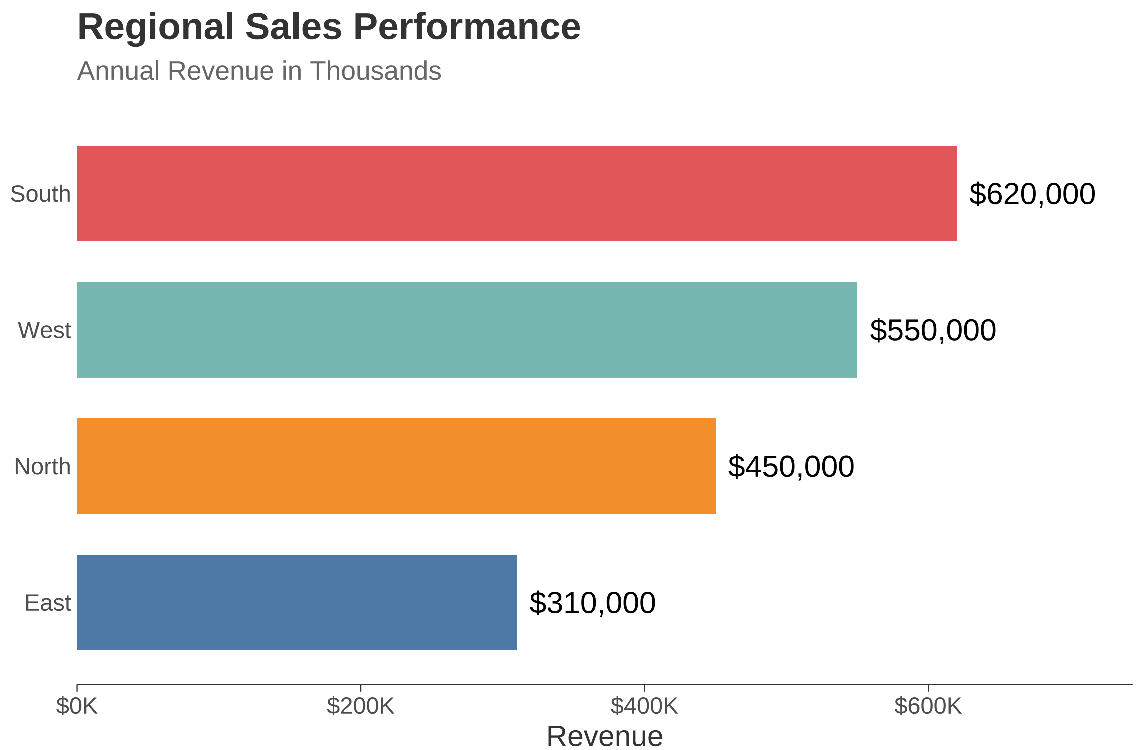


Figure 1: Revenue by Region

Line Plot

Line plots communicate change over time more clearly than any other chart type. Use them to show trends, the shape of change and to compare with other time lines.

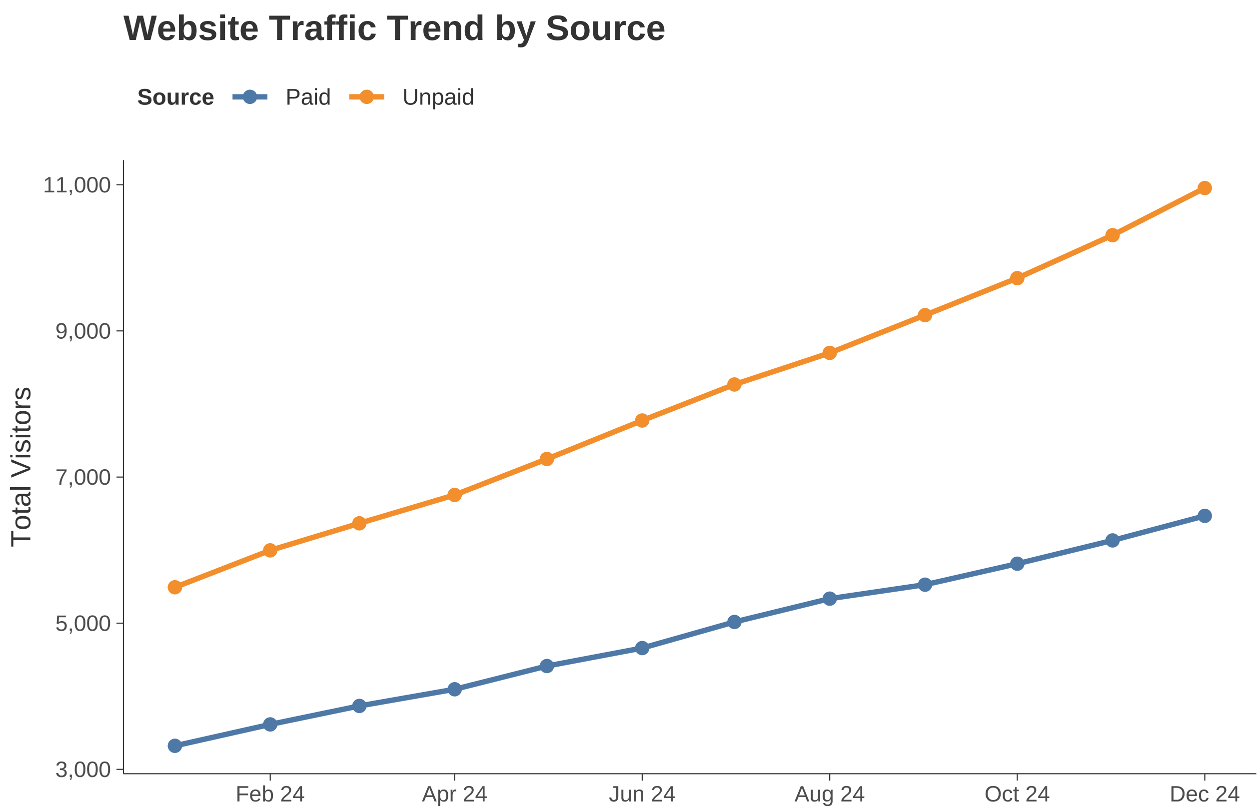


Figure 2: Monthly Website Traffic by Source

Scatterplot

This is the classic for visualizing the relationship between two continuous variables. Even more complex or modern graphics (like UMAP below)

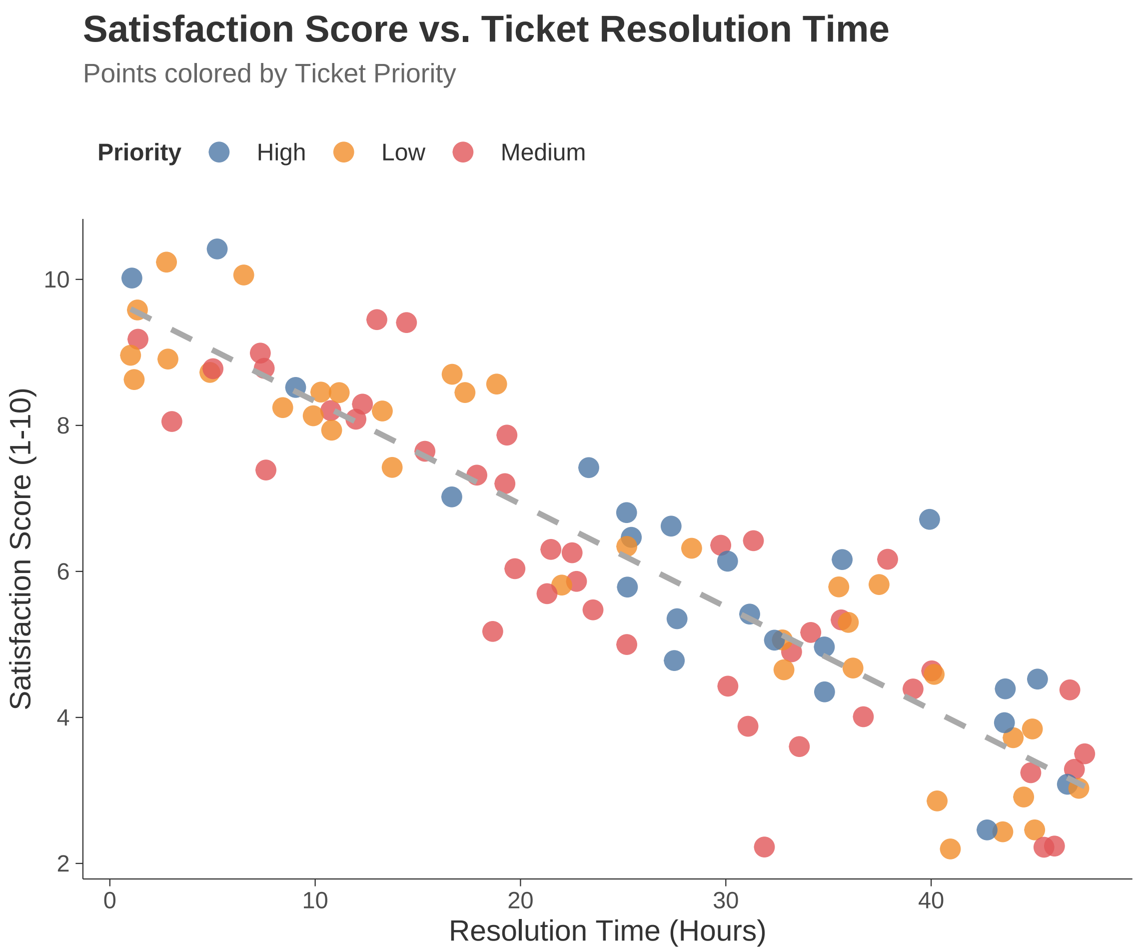


Figure 3: Customer Satisfaction vs. Ticket Resolution Time

Box plot

Boxplots reveal a lot of distributional information in a compact, standardized way and they are easy to compare. Typically combining categorical and continuous information in a single visual.

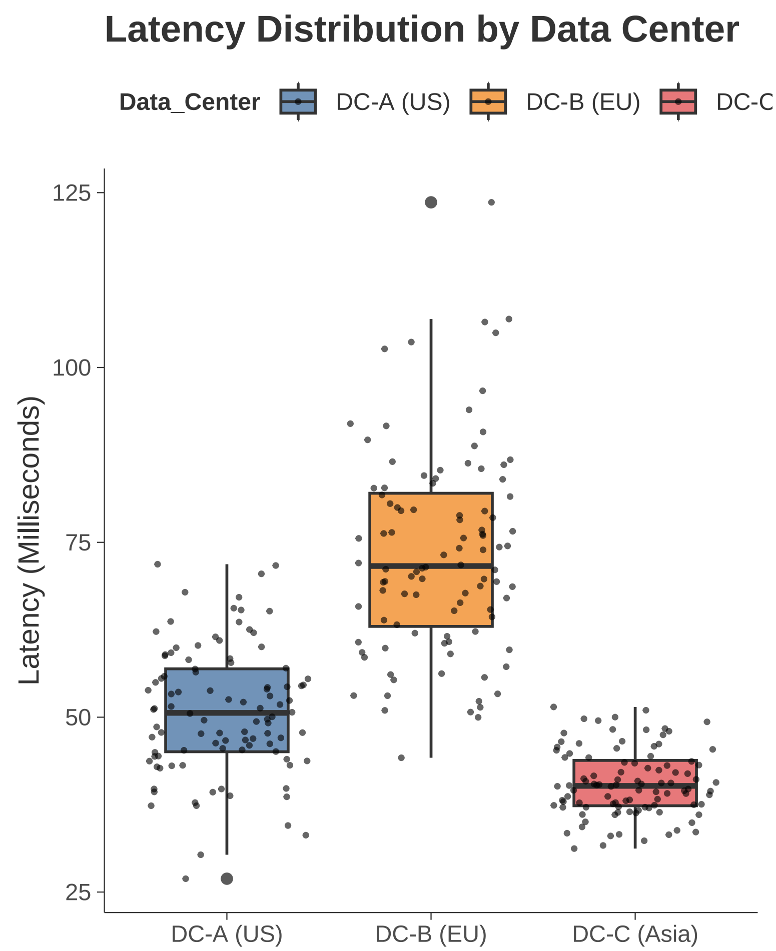


Figure 4: Distribution of Server Latency by Data Center

Mapping

We spent more time than you probably wanted on mapping, at its simplest, we can use various mapping libraries in R, this first one uses `rnaturalearth` but simpler ones that require fewer resources are like `maps`

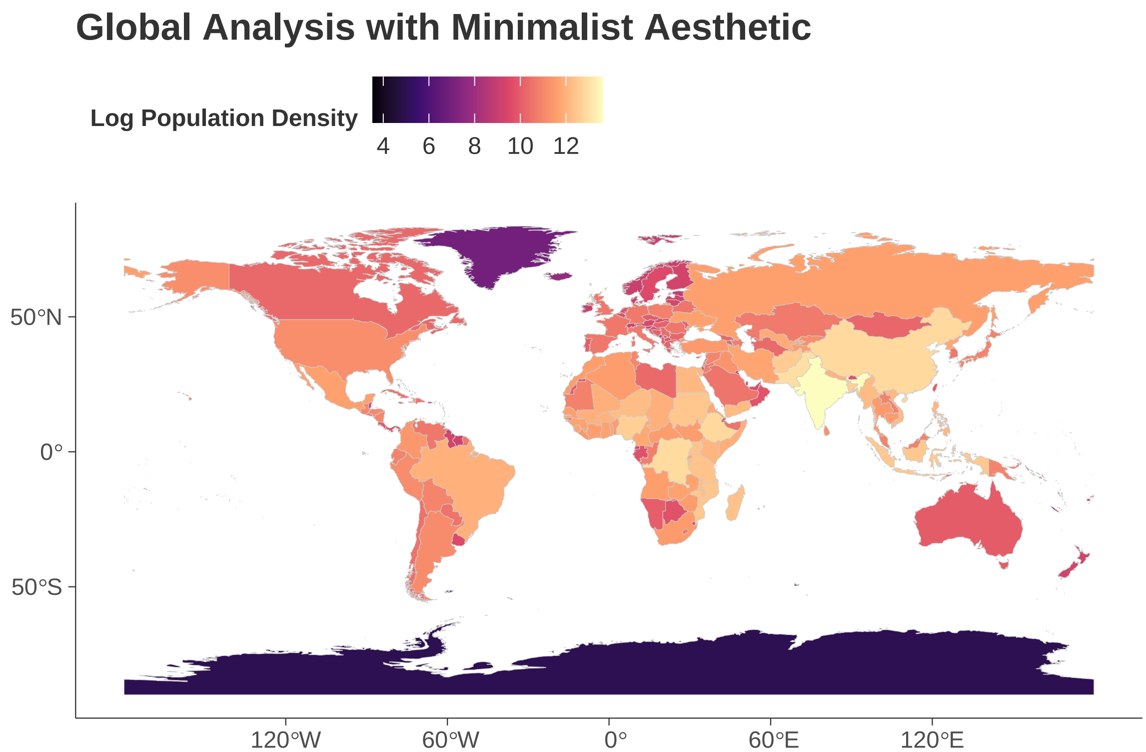


Figure 5: Themed Map of World Population Density

Heat Map

A slightly different map. Heatmaps are the go-to when your data are structured as a matrix so correlation, distance, confusion matrices, but also gene expression data

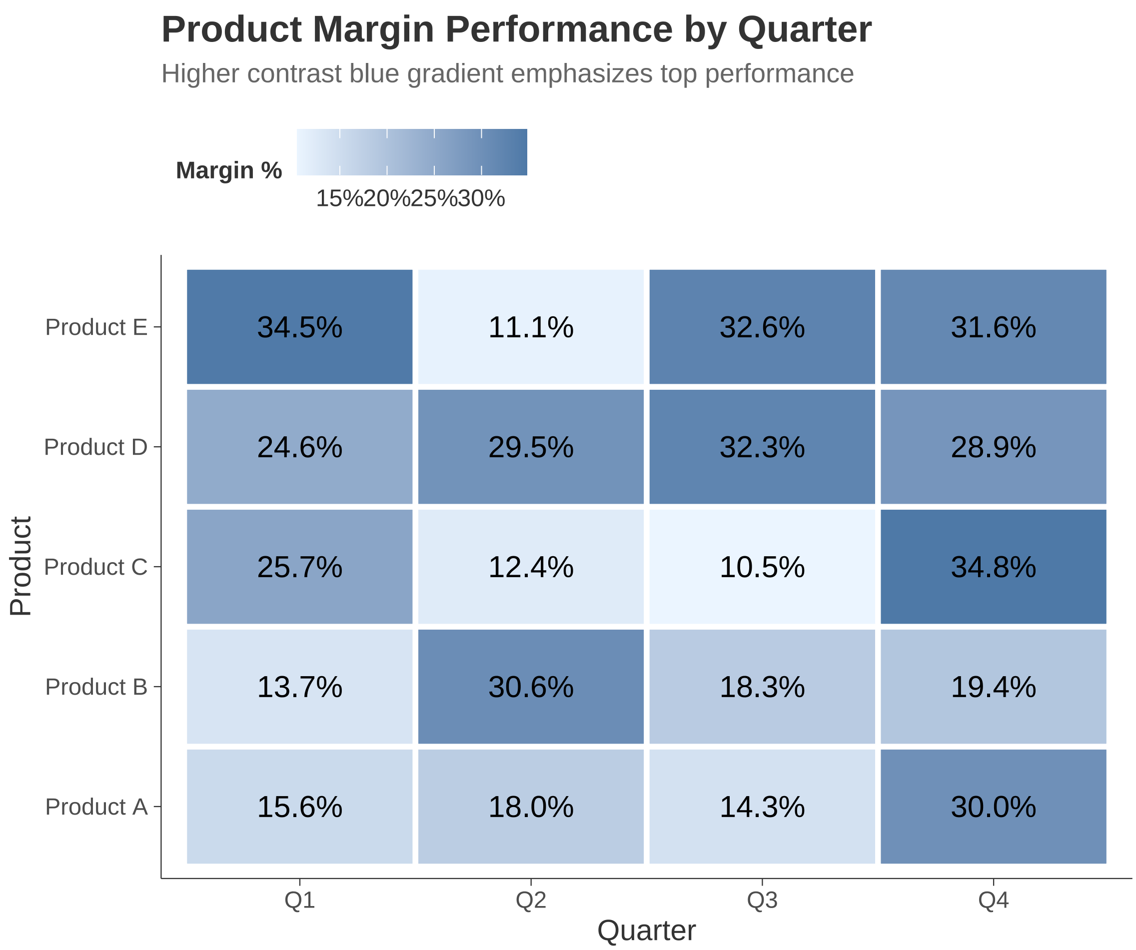


Figure 6: Quarterly Performance Heat Map (Tableau Sequential Color)

More Advanced/Newer

You do not typically see these in an undergraduate curriculum.

UMAP

UMAP (Uniform Manifold Approximation and Projection) is used for dimension reduction in data visualization and machine learning.

UMAP takes a dataset with many features (dimensions) and collapses into a 2D or 3D scatter plot. This allows us to identify clusters, patterns, and relationships that would be impossible to see in data with hundreds of columns.

UMAP areas of application

Genomics, Image Processing: Grouping similar images (e.g., separating photos of cats, dogs, and cars) based on pixel data. NLP – Visualizing word embeddings (grouping words with similar meanings together).

You can check out a cute video on UMAP: https://www.youtube.com/watch?v=eN0wFzBA4Sc&t=12s

UMAP1	UMAP2	Cluster
1.719762	2.393869	A
1.884911	2.384521	A
2.779354	2.166101	A
2.035254	1.495812	A
2.064644	1.940274	A
2.857533	1.859802	A

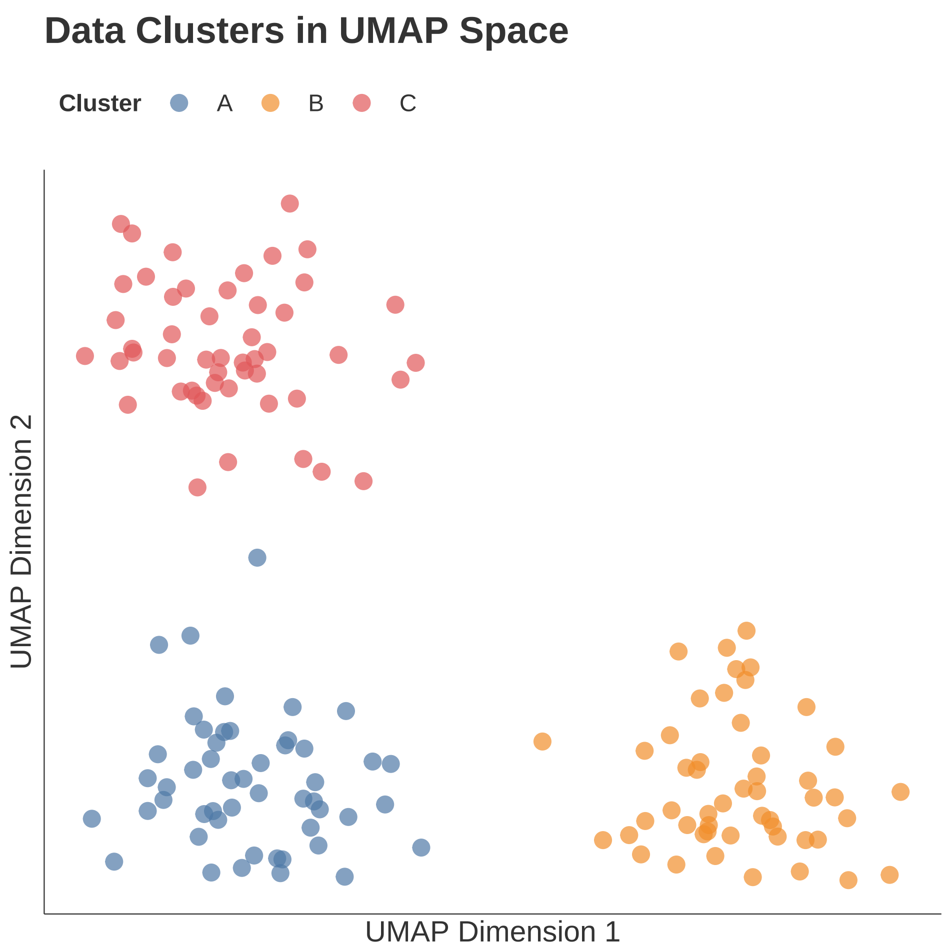
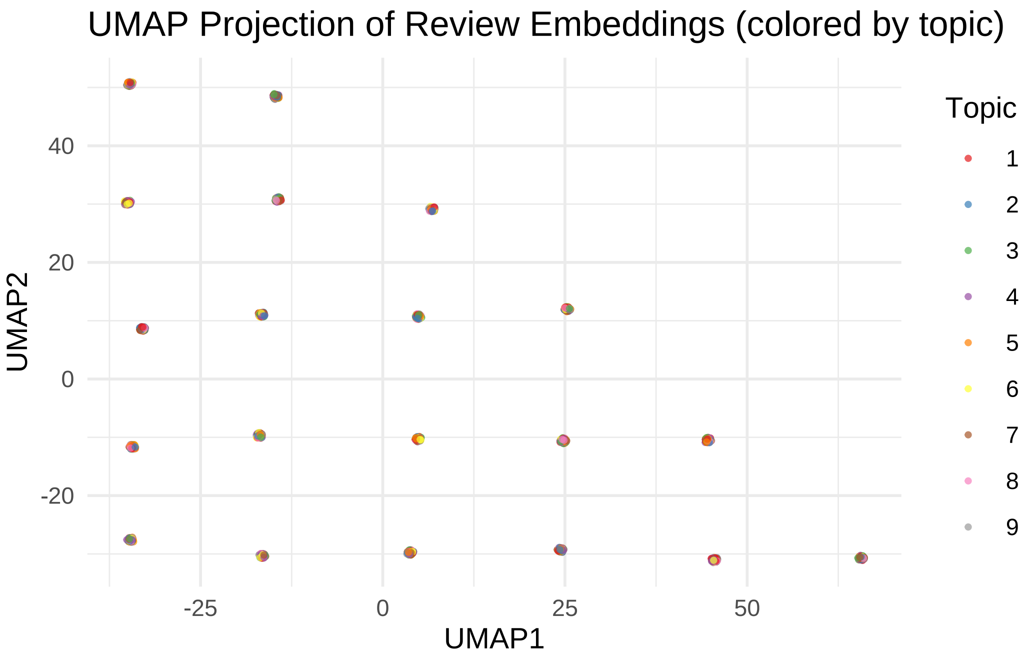


Figure 7: UMAP Projection of Simulated High-Dimensional Data

A Second UMAP Example

Slightly different data (a little large, you can view it separately)



Neural Net

The code below trains a neural network to classify iris flowers based on their measurements, and then it creates a visualization of that network's architecture and weights.

The line above uses the nnet package to build a neural network with a single hidden layer. Measurements from iris are pushed forward through 3 hidden processing units, and immediately outputs a classification, without ever looping back to reconsider or "remember" previous data. You can get a better idea see: https://www.lxt.ai/ai-glossary/neural-networks/

Visualize the network using the NeuralNetTools function plotnet

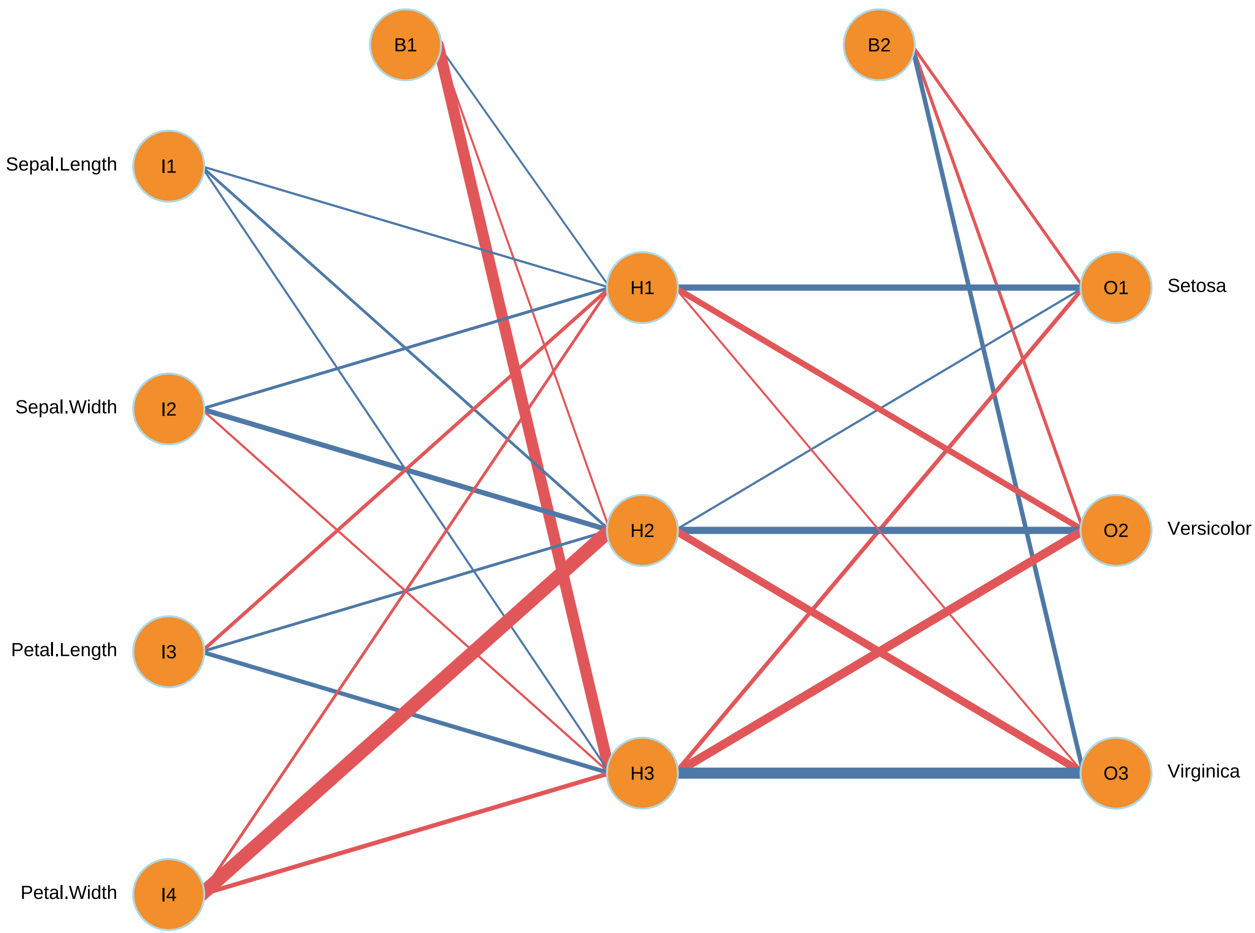


Figure 8: Neural Network Architecture

Network Data generally

We can find network data anywhere, the easiest form to use contains pairs which can serve as nodes. This is data from the Tax Treaties explorer (https://www.treaties.tax/en/)

Country A	Country B	Type	Date of signature	Effective
Australia	Kiribati	Original	1991	1991
Australia	Philippines	Original	1979	1980
Australia	Sri Lanka	Original	1989	1992
Argentina	Australia	Original	1999	2000
Australia	Samoa	Original	2009	2013
Australia	Marshall Islands	Original	2010	2012

Create Edge List

From this, we can create a data frame of edges (connections)

Source	Target	Status
Australia	Kiribati	In Force
Australia	Philippines	In Force
Australia	Sri Lanka	In Force
Argentina	Australia	In Force
Australia	Samoa	In Force
Australia	Marshall Islands	In Force

Create Node List

And also nodes:

id
Albania
Algeria
Angola
Argentina
Armenia
Australia

Get Some Geographic Coordinates

To make it mappable, we will need latitude and longitude. I found this list on internet:

longitude	latitude	COUNTRY	ISO	COUNTRYAFF	AFF_ISO
-170.7007	-14.30571	American Samoa	AS	United States	US
166.6380	19.30205	United States Minor Outlying Islands	UM	United States	US

longitude	latitude	COUNTRY	ISO	COUNTRYAFF	AFF_ISO
-159.7877	-21.22261	Cook Islands	CK	New Zealand	NZ
-149.4004	-17.67468	French Polynesia	PF	France	FR
-169.8688	-19.05231	Niue	NU	New Zealand	NZ
-128.3150	-24.36612	Pitcairn	PN	United Kingdom	GB

Aside - data cleaning

Join our nodes with the coordinates

Create A Graph Object

From the igraph library we can create a graph object, connecting our nodes and edges, relating size to the degree of connectedness and making it mappable:

Build a plot with ggraph

ggraph is the equivalent of ggplot() in ggplot2 but for our specific case. If you are comfortable with ggplot, it is effectively the same:

Global Tax Treaty Network

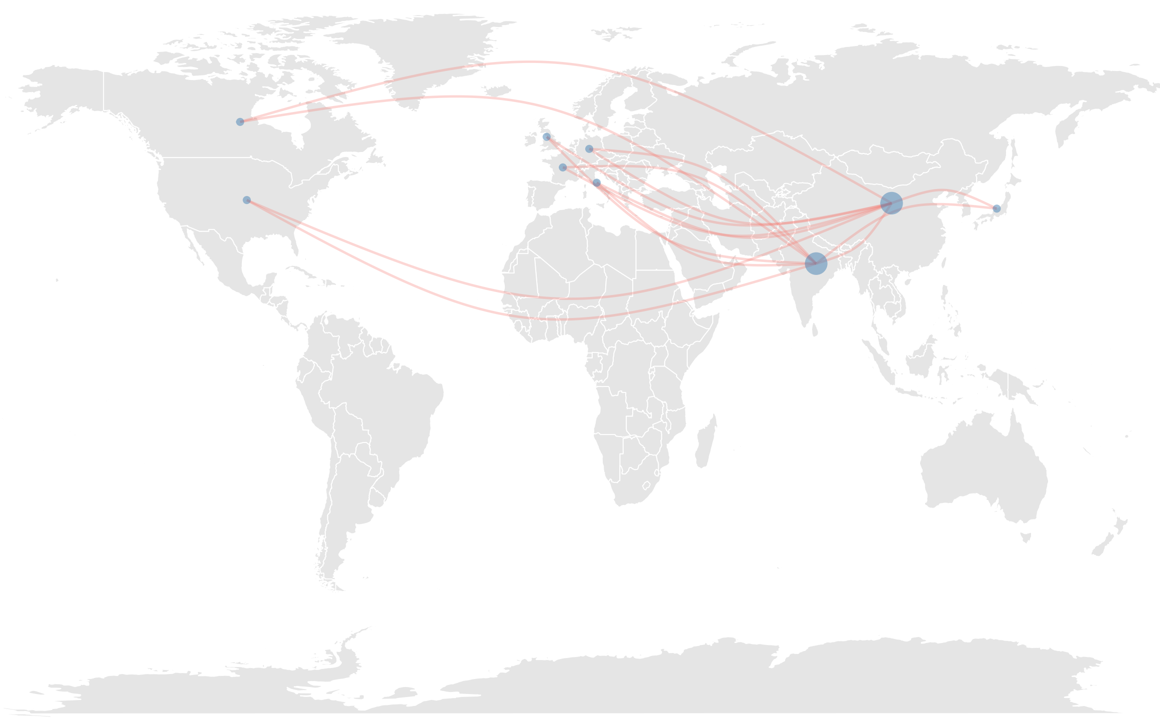


Figure 9: Geospatial visualization of treaty connections

DAG

What is DAG?

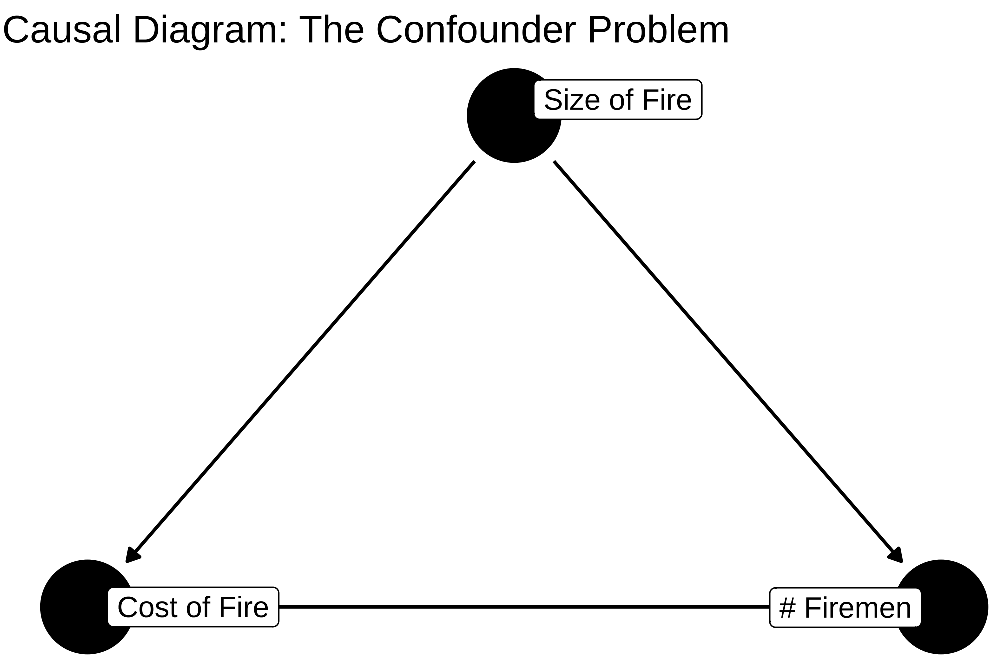
DAG means Directed Acyclic Graph. A DAG visualization is a graphical representation of a network where information or processes flow in a strictly one-way direction, with no loops.

- A DAG visualization is a graphical representation of a network where information or processes flow in a strictly one-way direction, with no loops.
- Used to represent pipelines (Data Engineering) and Causal Inference (statistics).

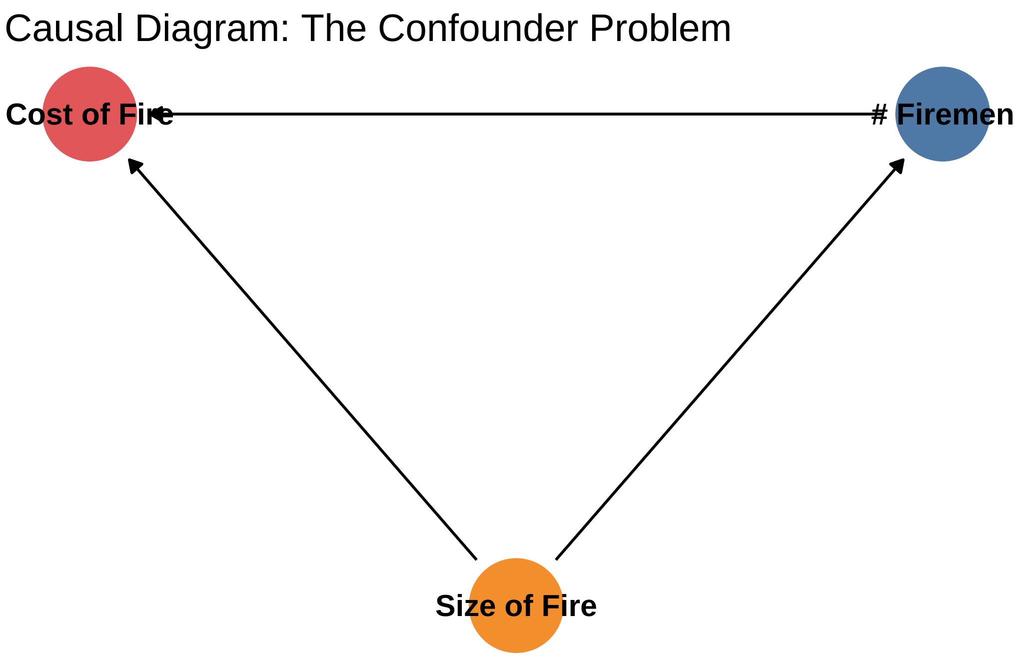
and DAG visualizations are

- Directed - All connections (edges) have a direction, usually shown with arrows ($4A \to B$)
- Acyclic - The arrows will never end up back where they started. There are no loops.
- Graph - It is a network of nodes (points) and edges (lines).

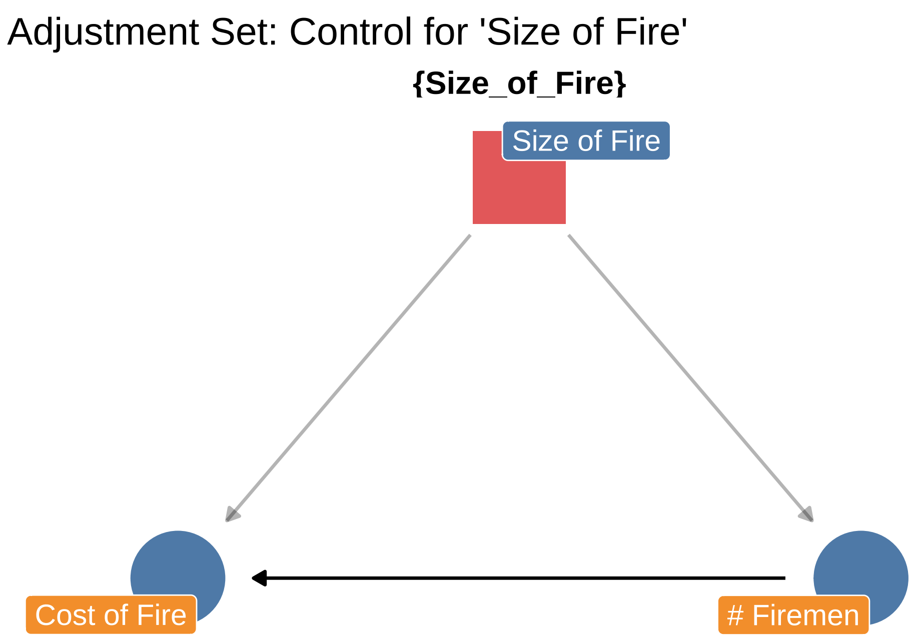
The "no loop" property makes DAG useful for visualizing dependencies and cause-and-effect.



ggdag makes ugly visualizations and they are a little tricky to modify. BUT we can convert it to something ggplot can use with tidy_dagitty in ggdag:



A little extra, the real use of DAG is to correct issues like the one we are seeing:



Takeaways

Theme and Palette

Generating consistent, publication-ready graphics in Quarto/R, regardless of the chart type requires theme and palette changes to get a certain style/look.

The use of Theme in ggplot2

Theme controls all non-data elements. For this report

- **Background:** White, minimal shading.
- **Gridlines:** Few or none
- **Axes:** Few ticks, high data-to-ink ratio (i.e., less ink spent on borders/ticks, more on the data itself).

Use ggplot functions like `theme_tufte()`, `theme_minimal()`, or `theme_light()`, followed by specific changes like removing minor gridlines or axis text to get the right look

The use of the Palette in ggplot2

The **Palette** controls the color mapping:

- Try using inclusive color schemes (e.g., colorblind)
- The `ggthemes` functions like `scale_color_tableau("Tableau 10")` uses the colors recognized as the Tableau standard.

The Visualization Bridge

Even for highly specialized research visualizations like **UMAP** or custom graphs, you typically generate the coordinates or structure first, and then you pipe that data structure into a **ggplot2** function OR use a package built on `ggplot2`, like `ggdag` or `geom_sf`.

The visualization is ultimately rendered by `ggplot2` and the theme and color scales are consistent. This is not perfect because DAG needed modification but UMAP was straightforward.